

EXPERIMENT 22

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Problem Statement

Write a menu-driven C program that accepts two positive integers **m** and **n**, where $m \leq n$. The menu must: (1) print multiplication tables of **m** and **n** side by side for multipliers 1 to 10, (2) display a multiplication grid for integers 1 to **m** and 1 to **n**, (3) for each integer from **m** to **n** print that integer followed by the next five consecutive integers, and (4) exit. The menu repeats until Exit is selected and invalid choices produce an error message.

Algorithm

1. Start.
2. Read positive integers **m** and **n**. Validate that $m > 0$, $n > 0$ and $m \leq n$; otherwise request them again.
3. Repeat the following menu until the user chooses 4:
 - 1 - Multiplication tables of **m** and **n** side by side
 - 2 - Multiplication grid from 1..**m** and 1..**n**
 - 3 - Consecutive-number rows from **m** through **n**
 - 4 - Exit
4. Read choice and use switch(choice).
5. For choice 1, loop **i** from 1 to 10 and print $m*i$ and $n*i$ on the same row.
6. For choice 2, use nested loops: rows $i=1..m$ and columns $j=1..n$; print $i*j$.
7. For choice 3, for each $i=m..n$, print **i**, **i+1**, **i+2**, **i+3**, **i+4**, **i+5**.
8. For choice 4, terminate the loop.
9. For any other choice, print an invalid-choice error.
10. Stop.

Test-Case Table

TC	Input / condition	Menu choice	Expected result
1	m=2, n=5	1	Tables of 2 and 5 printed side by side for 1..10.
2	m=2, n=4	2	2 x 4 multiplication grid containing $i*j$ values.
3	m=3, n=5	3	Rows: 3 4 5 6 7 8; 4 5 6 7 8 9; 5 6 7 8 9 10.
4	m=1, n=1	1	Boundary valid case; two identical tables of 1 are printed.
5	m=0, n=4 then m=2,n=4	-	Reject m=0; accept the next valid pair.
6	m=5, n=3 then m=3,n=5	-	Reject because $m > n$; accept the next valid pair.
7	m=-2, n=3 then m=1,n=3	-	Reject negative input; accept positive values.
8	m=2, n=3	9	Print "Invalid menu choice" and show menu again.
9	m=2, n=3	4	Exit program normally.

C Program

```
#include <stdio.h>

int main(void) {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Enter positive values with m <= n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\nMENU\n");
        printf("1. Multiplication tables of m and n side by side\n");
        printf("2. Multiplication grid\n");
        printf("3. Next five consecutive integers\n");
```

```

printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        printf("\nTable of %d\t\tTable of %d\n", m, n);
        for (i = 1; i <= 10; i++) {
            printf("%d x %d = %d\t\t%d x %d = %d\n",
                m, i, m*i, n, i, n*i);
        }
        break;

    case 2:
        printf("\nMultiplication Grid:\n\t");
        for (j = 1; j <= n; j++) printf("%4d", j);
        printf("\n");
        for (i = 1; i <= m; i++) {
            printf("%d\t", i);
            for (j = 1; j <= n; j++)
                printf("%4d", i*j);
            printf("\n");
        }
        break;

    case 3:
        printf("\nConsecutive integers:\n");
        for (i = m; i <= n; i++) {
            for (j = 0; j <= 5; j++)
                printf("%d ", i+j);
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid menu choice. Please enter 1 to 4.\n");
}
} while (choice != 4);

return 0;
}

```

Compilation and Execution

```

gcc experiment22.c -o experiment22
./experiment22

```

Sample Verification

Example: m = 2, n = 3, choice = 3.

Output rows are:

```

2 3 4 5 6 7
3 4 5 6 7 8

```

This confirms that each integer from m through n is followed by its next five consecutive integers.

Result

The menu-driven C program was prepared using loops and a switch statement. It validates m and n, performs all three required operations, handles invalid menu choices, and continues until the Exit option is selected.